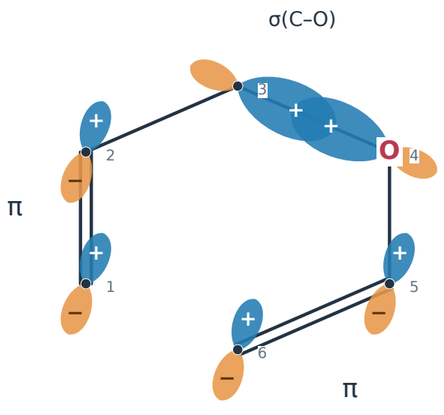
Claisen [3,3] | orbital phases along the reaction

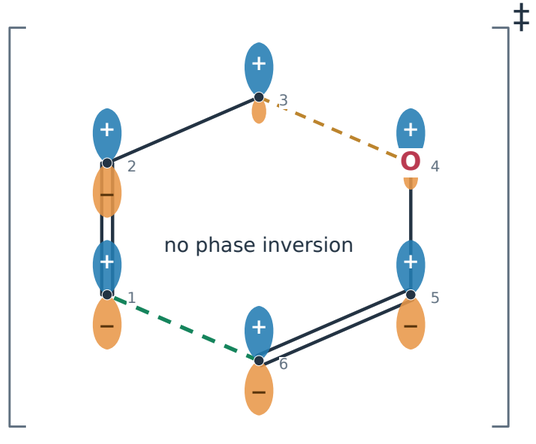
THERMALLY ALLOWED

suprafacial / suprafacial

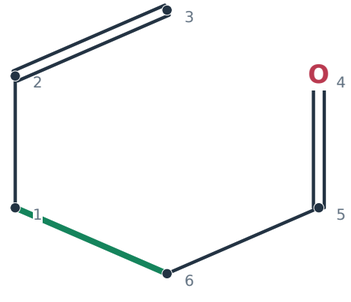
REACTANT ORBITALS	PROPOSED TRANSITION STATE	PRODUCT
Occupied π + σ + π bond orbitals	Hückel array · 6 electrons	Product connectivity



Δ



Δ



Pent-4-enal

Localized occupied orbitals; not one HOMO.

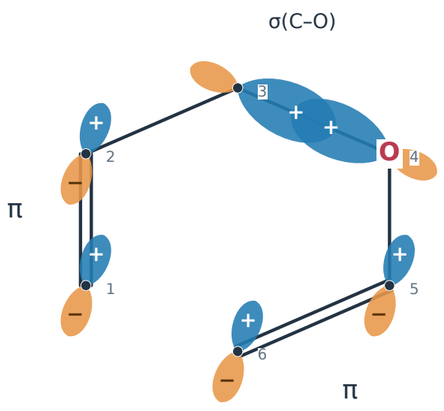
Local overlap basis; σ centers reorganize.

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / antarafacial

	Möbius array · 6 electrons	Proposed product connectivity



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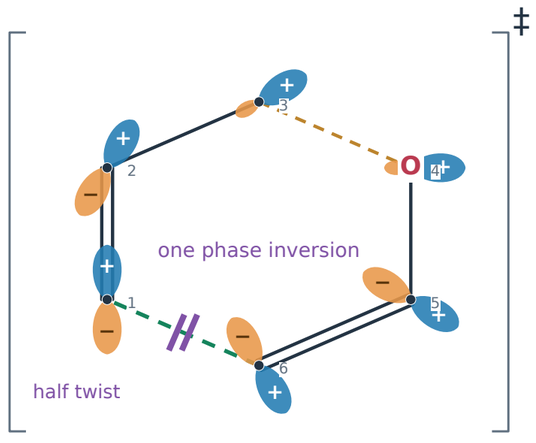

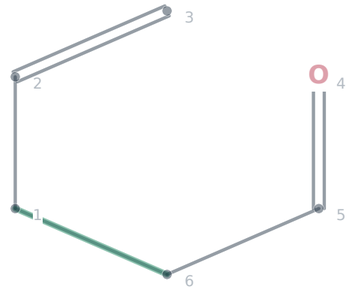

Pent-4-enal

Localized occupied orbitals; not one HOMO.

Local overlap basis; σ centers reorganize.

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

π/σ and transition-state local-basis drawings use different representations; neither is a calculated molecular HOMO.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.